Week 3 Assignment: Monte Carlo Simulation and Geometric Brownian Motion

Instructions

- Ensure that your code is well commented. AI may be used solely for debugging and technical assistance, not for generating the solution.
- Use Python language for simulation based questions.
- Submit both your code and type the written answers (very briefly) in the same file in Markdown.
- Submission of the first question is compulsory and must be done by 13th June EOD. The second question can be submitted by 16th June.

Task 1: Estimating Area Using Monte Carlo Simulation

Consider the quarter-circle of radius 1 lying in the first quadrant:

$$x^2 + y^2 = 1, \qquad x \geq 0,\ y \geq 0.$$

The quarter-circle is enclosed within the square

$$0 \leq x \leq 1, \qquad 0 \leq y \leq 1.$$

(a) Generate $N = 500$ random points uniformly inside the square.

(b) Determine whether each point lies inside or outside the quarter-circle.

(c) Let

$$N_{\text{inside}}$$

denote the number of points that fall inside the quarter-circle.

Using Monte Carlo simulation, estimate the area of the quarter-circle as

$$A_{\text{MC}} = \frac{N_{\text{inside}}}{N}.$$

(d) Using your estimate of the area, estimate the value of π.

(e) Repeat the experiment for

$$N = 500,\ 1000,\ 5000,\ 10000$$

and comment on how the accuracy changes as N increases.

(f) Plot the generated points, using different colors for points inside and outside the quarter-circle.

Task 2: Stock Price Simulation Using Geometric Brownian Motion

A stock price is assumed to follow Geometric Brownian Motion (GBM):

$$dS_t = \mu S_t\, dt + \sigma S_t\, dW_t.$$

The corresponding discrete-time model is

$$S_{t+1} = S_t \exp\left(\left(\mu - \frac{\sigma^2}{2}\right)\Delta t + \sigma\sqrt{\Delta t}\, Z_t\right),$$

where

$$Z_t \sim N(0,1).$$

Assume

$$S_0 = 100, \qquad \mu = 0.08, \qquad \sigma = 0.25, \qquad \Delta t = \frac{1}{252}.$$

(a) Explain the meaning of the parameters

$$S_t,\ \mu,\ \sigma,\ Z_t.$$

(b) Starting from $S_0 = 100$, generate one stock-price path using the above equation.

(c) Simulate the evolution of the stock price for 500 time steps.

(d) Plot the resulting stock-price path.

(e) Repeat the simulation 20 times and plot all paths on the same figure. (Try adding different color to each plot for a better looking plot)

(f) Compute the mean and standard deviation of the final stock price across the 20 simulations.

(g) Briefly explain why different simulations produce different future price paths even though the model parameters remain unchanged.

(h) Suppose you simulate 1000 independent GBM paths.

Estimate the probability that the stock price after 500 time steps exceeds 120:

$$P(S_{500} > 120).$$

(Hint: Use Monte Carlo Approach)
